Linear Algebra I Quiz 2 Solution

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If T is a linear transformation between two vector spaces, we use the notation $\operatorname{im} T := R(T)$, $\ker T := N(T)$.

1. (a) $\{(1,-1,0,0),(0,0,1,0),(0,0,0,1)\}$ is a basis for U
   $\{(1,0,0,0),(0,1,-1,0),(0,0,0,1)\}$ is a basis for W. $\dim U = \dim W = 3$.

   (b) $U \cap W = \{(a,b,c,d) \in F^4 \mid a+b=0, b+c=0\} = \{(a,b,c,d) \in F^4 \mid a+b=0, a=c\}$.
   $\{(1,-1,1,0),(0,0,0,1)\}$ is a basis for $U \cap W$, and $\dim U \cap W = 2$.
   $\dim(U+W) = \dim U + \dim W - \dim(U \cap W) = 3+3-2 = 4$.
   $U+W$ is not a direct sum since $U \cap W \neq \{0\}$.

2. (a) $T(1) = (1,1), T(x) = (0,1), T(x^2) = (0,1)$.

$$\begin{aligned}\ker T &= \{f = ax^2+bx+c \in F[x]_2 \mid f(0)=0, f(1)=0\} \\ &= \{ax^2+bx+c \in F[x]_2 \mid c=0, a+b+c=0\} \\ &= \{ax^2-ax \mid a \in F\}\end{aligned}$$

   Therefore, $\{x^2-x\}$ is a basis for $\ker T$ and nullity $T = \dim \ker T = 1$.

   (b)

$$\begin{aligned}\operatorname{im} T &= \{T(ax^2+bx+c) \mid a,b,c \in F\} \\ &= \{(c, a+b+c) \mid a,b,c \in F\} = F^2\end{aligned}$$

   since for any $(x,y) \in F^2$, pick $a = y-x, b = 0, c = x$, $a,b,c \in F$, we have $(c, a+b+c) = (x,y)$. Therefore, we can simply pick $\{(0,1),(1,0)\}$ as a basis for $\operatorname{im} T$ and rank $T = \dim \operatorname{im} T = 2$. T is onto but T is not one-to-one since $\ker T \neq \{0\}$.

3. (a) Let β be a basis for W, then β is a linear independent subset of V. Fix a basis γ for V, by replacement theorem, we can obtain a generating set δ of V containing β and $|\delta| = |\gamma| = \dim V$. Therefore, δ is also a basis for V.
   Write $\delta = \beta \cup \{x_1, \cdots, x_n\}$, and define $U = \operatorname{span}\{x_1, \cdots, x_n\}$, then from the fact that $\beta \cap \{x_1, \cdots, x_n\} = \emptyset$ and $\beta \cup \{x_1, \cdots, x_n\} = \delta$ being a basis for V, we have $V = W \oplus U$.

   (b) <u>P is well-defined</u>: For any $v \in V$, suppose $v = w+u = w'+u'$ for some $w, w' \in W, u, u' \in U$, then $w - w' = u - u' \in W \cap U = \{0\}$. That is, $w = w', u = u'$, and $P(w+u) = w = w' = P(w'+u')$, so P is well-defined.
   <u>P is linear</u>: Given $v_1, v_2 \in V$, v_1, v_2 can be written as $v_1 = w_1+u_1$, $v_2 = w_2+u_2$ where $w_1, w_2 \in W, u_1, u_2 \in U$. Then $v_1+v_2 = (w_1+w_2)+(u_1+u_2)$ with $w_1+w_2 \in W, u_1+u_2 \in U$, so $P(v_1+v_2) = w_1+w_2 = P(v_1)+P(v_2)$. Given any $a \in F, v \in V$, decompose $v = w+u$ for some $w \in W, u \in U$, then

$av = a(w+u) = aw + au$ with $aw \in W, au \in U$, so $P(av) = aw = aP(v)$. Thus, P is linear.

$\underline{\text{im}\, P = W}$: For any $x \in \text{im}\, P$, $x = P(v)$ for some $v \in V$, write $v = w + u$ for some $w \in W, u \in U$, then $x = P(v) = w \in W$. $\text{im}\, P \subseteq W$. On the other hand, given $w \in W$, it is clear that $P(w) = w$ since $w = w + 0$ with $w \in W, 0 \in U$. Therefore $W \subseteq \text{im}\, P$, so $W = \text{im}\, P$.

$\underline{\ker P = U}$: Given $u \in U$, $u = 0 + u$ where $0 \in W, u \in U$, so $P(u) = 0$, i.e. $u \in \ker P$, so $U \subseteq \ker P$. On the other hand, if $y \in \ker P$, then $P(y) = 0$, decompose $y = w + z$ for some $w \in W, z \in U$, we have $P(y) = w = 0$, thus $y = 0 + z \in U$, so $\ker P \subseteq U$. $P = \ker U$.